Review plan · for the lecturer

Teaching vision and generation: unit check

University

Time
45 min

Materials
Paper; The four lessons

Learning goals

- Four lessons
- One sitting

Activity: Work it

1. Answer all ten
2. Then read each explain

Discussion

- Which lesson was the miss?

Exit ticket

- Name the idea you will redo.

Try it in your next lecture
Redo the one item you missed, on paper, before the next course.

Quiz sheet · for students

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Name _____ Group _____ Date _____

1 **The lab's Upright-edges filter sits on this patch (1 is bright, 0 is dark). What number does it write?**

```
patch = [[0, 0, 1],
         [0, 0, 1],
         [0, 0, 1]]
edges = [[-1, 0, 1],
         [-2, 0, 2],
         [-1, 0, 1]]
```

◯ 3 ◯ 0 ◯ 4

2 **In a trained vision network, the nine numbers of each filter are learned from data.**

◯ True ◯ False

3 **Match each part of a vision block to what it throws away.**

Convolution	• • • •	Exactly where in each block the pattern was
ReLU	• • • •	Every pattern but the one its filter looks for
Max pooling	• • • •	Negative responses, set to 0

4 **The detector's scores: close-set face 0.894, wide face 0.548, box 0. The line is 0.55. Which result is a miss?**

◯ The close-set face ◯ The box ◯ The wide face

5 **A detector that is right on two of its three test drawings can still miss every face of one group.**

◯ True ◯ False

6 **Match each result to its name.**

A face scored below the line	• • • •	A false alarm
A box scored above the line	• • • •	A correct rejection
A box scored below the line	• • • •	A miss

7 **Why does a random code decode to a plausible picture more often in a variational autoencoder than in a plain one?**

◯ Its code has more numbers than the picture ◯ It stores every training picture

◯ Its loss keeps the codes in one smooth, shared cloud

8 **In the lab, leaving the code where it starts gives back the original picture exactly.**

◯ True ◯ False

9 **Match each picture's story to the method that most likely made it.**

A blurry, average-looking rebuild of a face it was given	• • • •	Diffusion
Sharp pictures, but nearly every sample looks the same	• • • •	Autoencoder
Started from pure noise and cleaned in many small steps	• • • •	GAN

10 **What is diffusion's main practical cost?**

◯ It needs a second network to judge real or fake

◯ Each sample repeats the denoising step many times, so it is slow

◯ It can only rebuild pictures it was given

Answers · for the lecturer

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1. **4**
   Only the right column is bright: 1×1 + 1×2 + 1×1 = 4, and the other six products are 0.

2. **True**
   Training sets the filter numbers; nobody picks them.

3. **Convolution — Every pattern but the one its filter looks for**
   ReLU — Negative responses, set to 0
   Max pooling — Exactly where in each block the pattern was
   Each part keeps less than it receives, so later layers can combine edges into shapes without every pixel.

4. **The wide face**
   A miss is a real face below the line: 0.548 is under 0.55.

5. **True**
   Its only miss is the wide face: 0% missed for close-set faces, 100% for wide ones.

6. **A face scored below the line — A miss**
   A box scored above the line — A false alarm
   A box scored below the line — A correct rejection
   Which mistake costs more depends on the use; a face tool that misses one group is not ready, however good its average.

7. **Its loss keeps the codes in one smooth, shared cloud**
   A VAE encodes each picture as a small cloud and pulls all the clouds toward one shared cloud, so codes between pictures decode to something sensible; a plain autoencoder can leave gaps.

8. **False**
   The difference is 0.025, not 0: the blob is in the right place, but the two bright specks are gone, because two numbers cannot hold them.

9. **A blurry, average-looking rebuild of a face it was given — Autoencoder**
   Sharp pictures, but nearly every sample looks the same — GAN
   Started from pure noise and cleaned in many small steps — Diffusion
   Each method has its own loss and typical failure: blur, collapse to a few samples, or slow many-step sampling.

10. **Each sample repeats the denoising step many times, so it is slow**
    Generation repeats the learned step tens to a thousand times.